3.1.3

(1) $I_B = I_c/\beta = 1.5/80\ mA = 18.75\mu A$.

$I_{RC} = I_B + I_c = (1.5+18.75\times10^{-3})\ mA = 1.51875 mA$.

$U_{CB} = U_{CE} - U_{BE} = (6-0.7)V = 5.3V$.

$\therefore R_B = U_{CB}/I_B = 283k\Omega$.

$R_c = (U_{cc} - U_{CE})/I_{RC} = 12/1.51875\ k\Omega = 7.9k\Omega$.

(2)
$$\begin{cases} I_{RC}\cdot R_C + U_{CE} = U_{cc} \\ U_{CE} = I_B\cdot R_B + U_{BE} \\ I_B = I_c/\beta \\ I_c + I_B = I_{RC} \end{cases} \Rightarrow \begin{matrix} I_c = 1.214mA \\ U_{CE} = 5.71V \end{matrix}$$

3.14.

(a) 是饱和失真，可通过增大R_B或减小R_C消除。

(b) 是截止失真，可通过减小R_B实现

(c) 是两者都有，需减小U_i

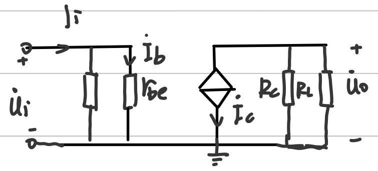
3.15

(1) $I_B = \frac{U_{cc}-U_{BE}}{R_B} = \frac{12-0.7}{280\times10^3}A = 40\mu A$.

$I_c = \beta\cdot I_B = 50\times40\times10^{-6}A = 2mA$.

$U_{CE} = U_{cc} - I_c R_c = 12-2\times3 = 6V$.

(2)



$r_{be} = 200 + \frac{26}{I_E}(1+\beta)$

$I_E = (1+\beta)I_B$

$\therefore r_{be} = 850\Omega$.

$U_o = I_c\cdot\frac{R_c\cdot R_L}{R_c+R_L}$.

$U_i = I_b\cdot r_{be}$

$A_u = -\frac{U_o}{U_i} = -88.2$

$r_i = \frac{U_i}{I_i} = U_i/[I_b(1+\frac{r_{be}}{R_B})] = 0.847k\Omega$.

$r_o = R_c = 3k\Omega$.

(3) $I_c = \frac{U_{cc}-U_{CE}}{R_c} = 2.5mA$

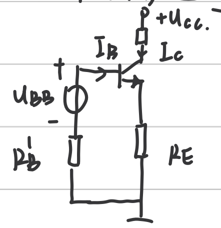
$I_b = I_c/\beta = 50\mu A$ $\quad R_B = \frac{U_{cc}-U_{BE}}{I_b} = 226k\Omega$.

(4) $I_b = I_c/\beta = 30\mu A$

$R_c = \frac{U_{cc}-U_{CE}}{I_c} = 4k\Omega$ $\quad R_B = \frac{U_{cc}-U_{BE}}{I_b} = 376.7k\Omega$

3.1.6

(1) 对电路进行戴维宁等效.



$$U_{BB} = U_{CC}\frac{R_{B1}R_{B2}}{R_{B1}+R_{B2}} = 3V$$

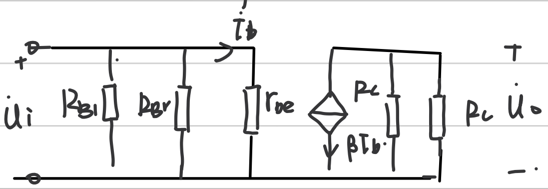
$$R_B' = \frac{R_{B1}R_{B2}}{R_{B1}+R_{B2}} = 7.5k\Omega.$$

$$U_{BB} = U_{BE} + I_E R_E + I_B R_B' \qquad I_E = (1+\beta)I_B$$

$$\therefore I_B = 0.0192\,mA \qquad I_C = \beta I_B = 0.96\,mA.$$

$$U_{CE} = U_{CC} - I_C R_C - I_E R_E = 4.95V.$$

(2) 画出微变等效模型:



$$r_{be} = 200 + \frac{26}{I_E}(1+\beta)$$

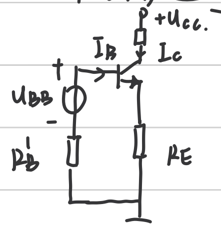
$$A_u = -\frac{U_o}{U_i} = -\frac{I_c\frac{R_L R_C}{R_L+R_C}}{I_b r_{be}} = -64.5$$

(3) $r_i = \left(\frac{1}{R_{B1}} + \frac{1}{R_{B2}} + \frac{1}{r_{be}}\right)^{-1} = 1.29\,k\Omega.$

$r_o = R_C = 5.1\,k\Omega.$

3.1.7

(1) 对电路进行戴维宁等效.



$$U_{BB} = U_{CC}\frac{R_{B1}R_{B2}}{R_{B1}+R_{B2}} = 3V$$

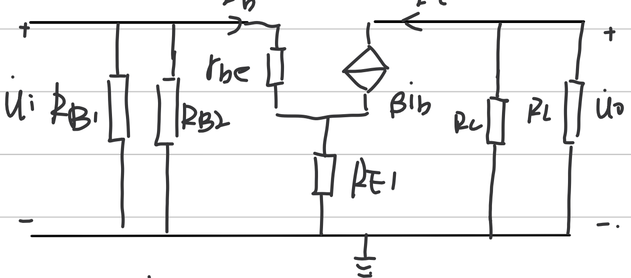
$$R_B' = \frac{R_{B1}R_{B2}}{R_{B1}+R_{B2}} = 7.5k\Omega.$$

$$U_{BB} = U_{BE} + I_E R_E + I_B R_B' \qquad I_E = (1+\beta)I_B$$

$$\therefore I_B = 0.0192\,mA \qquad I_C = \beta I_B = 0.96\,mA.$$

$$U_{CE} = U_{CC} - I_C R_C - I_E R_E = 4.95V.$$

(2) 画出微变等效电路.



$$r_{be} = 200 + (1+\beta)\frac{26}{I_E}$$

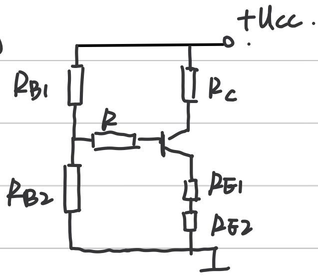
$$A_u = -\frac{U_o}{U_i} = \frac{-I_c\cdot\frac{R_L R_C}{R_C+R_L}}{I_B r_{be} + (1+\beta)I_B R_{E1}} = -8.5.$$

(3) $r_i = \left(\frac{1}{R_{B1}} + \frac{1}{R_{B2}} + \frac{1}{r_{be}+(1+\beta)R_{E1}}\right)^{-1} = 4.58\,k\Omega.$

$r_o = R_C = 5.1\,k\Omega.$

3.1.8.

(1)



T放截面等效.

$$R_B' = R + \frac{R_{B1}R_{B2}}{R_{B1}+R_{B2}} = 156\,k\Omega$$

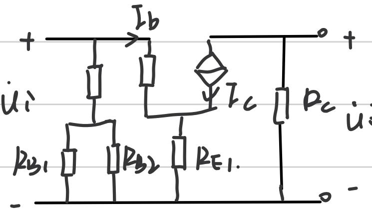
$$U_{BB} = \frac{R_{B2}}{R_{B1}+R_{B2}} U_{CC} = 4.8V$$

$$I_B = \frac{U_{BB}-U_{BE}}{R_B' + (R_{E1}+R_{E2})(1+\beta)} = 0.0159\,mA$$

$$I_C = \beta \cdot I_B = 0.795\,mA. \qquad I_E = (1+\beta) I_B = 0.81\,mA.$$

$$U_{CE} = U_{CC} - I_C R_C - I_E(R_{E1}+R_{E2}) = 6.33\,V$$

(2)



$$A_u = -\frac{u_o}{u_i} = \frac{-I_c R_c}{r_{be}+(1+\beta)R_{E1}} = -36.8.$$

$$r_i = R + \frac{R_{B1}R_{B2}}{R_{B1}+R_{B2}} = 6.64\,k\Omega. \qquad r_o = R_c = 5.1\,k\Omega.$$

3.1.9.

放截面等效.

$$U_{BB} = \frac{R_{B2}}{R_{B1}+R_{B2}} U_{CC} = \frac{22}{43+22} \times 15V = 5.077V.$$

$$R_B = \frac{R_{B1}R_{B2}}{R_{B1}+R_{B2}} = 14.55\,k\Omega.$$

$$I_B = \frac{U_{BB}-U_{BE}-U_Z}{R_B} = 0.0946\,mA. \qquad I_C = \beta I_B = 5.676\,mA.$$

$$U_{CE} = U_{CC} - I_C R_C - U_Z = 6.324\,V.$$

(2) $$A_u = -\frac{u_o}{u_i} = \frac{-I_c R_c}{I_b r_{be}} = -126.4.$$

$$r_i = \left(\frac{1}{R_{B1}} + \frac{1}{R_{B2}} + \frac{1}{r_{be}}\right)^{-1} = 0.46\,k\Omega.$$

$$r_o = R_c = 1\,k\Omega.$$